

CO4_AT2 CASE Study

Question-1:

Write a C++ program to calculate the percentage of a student using multi-level inheritance. Accept the marks of three subjects in base class. A class will be derived from the above mentioned class which includes a function to find the total marks obtained and another class derived from this class which calculates and displays the percentage of student.

```
#include <iostream>
using namespace std;

class Student
{
protected:
    int m1, m2, m3;

public:
    void accept()
    {
        cout << "Enter marks of 3 subjects: ";
        cin >> m1 >> m2 >> m3;
    }
};

class Total : public Student
{
protected:
    int total;

public:
    void findTotal()
    {
        total = m1 + m2 + m3;
    }
};

class Percentage : public Total
{
    float percentage;

public:
    void calculate()
    {
        percentage = total / 3.0;
    }

    void display()
    {
        cout << "Total Marks: " << total << endl;
        cout << "Percentage: " << percentage << "%" << endl;
    }
};

int main()
{
```

```

    Percentage p;

    p.accept();
    p.findTotal();
    p.calculate();
    p.display();

    return 0;
}

```

Question-2:

Design a C++ program using hierarchical inheritance with a base class Person containing the data members name, address, and phone_no. Derive two classes, Employee and Manager, from the class Person. The Employee class should contain eno and ename, while the Manager class should contain designation, department_name, and basic_salary.

Write a menu-driven C++ program to:

- Accept all details of n employees and n managers.
- Display the details of all employees and managers.
- Find and display the manager who has the highest basic salary.
- Exit the program.

```

#include <iostream>
#include <string>
using namespace std;

class Person
{
protected:
    string name, address;
    long phone_no;

public:
    void acceptPerson()
    {
        cout << "Enter name: ";
        cin >> name;

        cout << "Enter address: ";
        cin >> address;

        cout << "Enter phone number: ";
        cin >> phone_no;
    }
};

class Employee : public Person
{
    int eno;
    string ename;

public:
    void accept()
    {
        acceptPerson();
    }
};

```

```

        cout << "Enter employee number: ";
        cin >> eno;

        cout << "Enter employee name: ";
        cin >> ename;
    }

    void display()
    {
        cout << "\nEmployee Details\n";
        cout << "Name: " << name << endl;
        cout << "Address: " << address << endl;
        cout << "Phone: " << phone_no << endl;
        cout << "Employee No: " << eno << endl;
        cout << "Employee Name: " << ename << endl;
    }
};

class Manager : public Person
{
    string designation, department_name;
    double basic_salary;

public:
    void accept()
    {
        acceptPerson();

        cout << "Enter designation: ";
        cin >> designation;

        cout << "Enter department: ";
        cin >> department_name;

        cout << "Enter basic salary: ";
        cin >> basic_salary;
    }

    void display()
    {
        cout << "\nManager Details\n";
        cout << "Name: " << name << endl;
        cout << "Address: " << address << endl;
        cout << "Phone: " << phone_no << endl;
        cout << "Designation: " << designation << endl;
        cout << "Department: " << department_name << endl;
        cout << "Basic Salary: " << basic_salary << endl;
    }

    double getSalary()
    {
        return basic_salary;
    }

    string getName()
    {
        return name;
    }
};

```

```

    }
};

int main()
{
    int n;

    cout << "Enter number of employees: ";
    cin >> n;

    Employee e[10];
    Manager m[10];

    for (int i = 0; i < n; i++)
    {
        cout << "\nEnter Employee " << i + 1 << " details\n";
        e[i].accept();
    }

    for (int i = 0; i < n; i++)
    {
        cout << "\nEnter Manager " << i + 1 << " details\n";
        m[i].accept();
    }

    int choice;

    do
    {
        cout << "\n1. Display Employees";
        cout << "\n2. Display Managers";
        cout << "\n3. Highest Salary Manager";
        cout << "\n4. Exit";
        cout << "\nEnter choice: ";
        cin >> choice;

        switch (choice)
        {
            case 1:
                for (int i = 0; i < n; i++)
                    e[i].display();
                break;

            case 2:
                for (int i = 0; i < n; i++)
                    m[i].display();
                break;

            case 3:
            {
                int pos = 0;

                for (int i = 1; i < n; i++)
                {
                    if (m[i].getSalary() > m[pos].getSalary())
                        pos = i;
                }
            }
        }
    }
}

```

```

        cout << "\nManager with Highest Salary: "
            << m[pos].getName() << endl;
        cout << "Salary: " << m[pos].getSalary() << endl;
        break;
    }

    case 4:
        cout << "Program ended";
        break;

    default:
        cout << "Invalid choice";
    }

} while (choice != 4);

return 0;
}

```

Question-3:

1. Consider two base classes

worker(int code, char name, float salary), officer(float DA, HRA)

class manger(float TA(is 10% of salary), gross sal) is derived from both base classes. Write necessary member function.

```

#include <iostream>
#include <string>
using namespace std;

class Worker
{
protected:
    int code;
    string name;
    float salary;

public:
    void acceptWorker()
    {
        cout << "Enter code: ";
        cin >> code;

        cout << "Enter name: ";
        cin >> name;

        cout << "Enter salary: ";
        cin >> salary;
    }
};

class Officer
{
protected:
    float DA, HRA;

```

```

public:
    void acceptOfficer()
    {
        cout << "Enter DA: ";
        cin >> DA;

        cout << "Enter HRA: ";
        cin >> HRA;
    }
};

class Manager : public Worker, public Officer
{
    float TA;
    float gross_sal;

public:
    void calculate()
    {
        TA = salary * 0.10;
        gross_sal = salary + DA + HRA + TA;
    }

    void display()
    {
        cout << "\nCode: " << code << endl;
        cout << "Name: " << name << endl;
        cout << "Salary: " << salary << endl;
        cout << "DA: " << DA << endl;
        cout << "HRA: " << HRA << endl;
        cout << "TA: " << TA << endl;
        cout << "Gross Salary: " << gross_sal << endl;
    }
};

int main()
{
    Manager m;

    m.acceptWorker();
    m.acceptOfficer();
    m.calculate();
    m.display();

    return 0;
}

```

Question-4:

Write a C++ program to define a base class Item (item-no, name, price). Derive a class

Discounted-Item (discount-percent). A customer purchases 'n' items. Display the item-wise bill and total amount using appropriate format.

```
#include <iostream>
#include <string>
using namespace std;

class Item
{
protected:
    int item_no;
    string name;
    float price;

public:
    void acceptItem()
    {
        cout << "Enter item number: ";
        cin >> item_no;

        cout << "Enter item name: ";
        cin >> name;

        cout << "Enter price: ";
        cin >> price;
    }
};

class DiscountedItem : public Item
{
    float discount_percent;
    float final_price;

public:
    void accept()
    {
        acceptItem();

        cout << "Enter discount percentage: ";
        cin >> discount_percent;
    }

    void calculate()
    {
        final_price = price - (price * discount_percent / 100);
    }

    void display()
    {
        cout << "\nItem No: " << item_no << endl;
        cout << "Name: " << name << endl;
        cout << "Original Price: " << price << endl;
        cout << "Discount: " << discount_percent << "%" << endl;
        cout << "Final Price: " << final_price << endl;
    }

    float getFinalPrice()
```

```

    {
        return final_price;
    }
};

int main()
{
    int n;
    float total = 0;

    cout << "Enter number of items: ";
    cin >> n;

    DiscountedItem d[10];

    for (int i = 0; i < n; i++)
    {
        cout << "\nEnter Item " << i + 1 << " details\n";

        d[i].accept();
        d[i].calculate();
    }

    cout << "\n----- ITEM WISE BILL ----- \n";

    for (int i = 0; i < n; i++)
    {
        d[i].display();
        total = total + d[i].getFinalPrice();
    }

    cout << "\nTotal Amount: " << total << endl;

    return 0;
}

```